

AI VisionGuard

Introduction to Computer Vision:HelmNet PGP-AIML-BA-UTA-Mar25-E

Date 10/05/2025

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Executive Summary

Challenge & Solution

Manual monitoring of worker safety compliance (specifically helmet use) is inefficient and error-prone at scale, risking severe injury or fatality. We have developed an **AI-based Computer Vision solution** to automate site analysis, rapidly flagging non-compliance to enhance workplace safety.

Model Performance & Strategy

- **Models Evaluated:** The feasibility study compared the performance of four different model architectures:
 - **CNN Base Model** (Trained from scratch)
 - **VGG16 Base Model** (Transfer learning, no fine-tuning)
 - **VGG16 + FFNN Model** (Transfer learning with additional layers)
 - **VGG16 + FFNN with Data Augmentation**
- **Best Model:** The **VGG-16 model with a Feed-Forward Neural Network (FFNN) and Data Augmentation** proved to be the most robust approach for the small dataset.
- **Metric Prioritization:** Due to the high-risk nature of the task, **Recall metrics were prioritized** to ensure the system minimizes False Negatives (missed violations). The best model achieved **1.0 (perfect) scores** on the small dataset, which, while statistically successful, warrants caution against potential overfitting in the PoC phase.
- **Complexity Observation:** Classification proved to be **quite complex** due to varied job site environmental factors, confirming the need for advanced models and data augmentation.

Executive Summary

Key Recommendations for Scale-Up

1. **Data Diversity is Critical:** Immediate action is needed to train the model on a **larger, more diverse image dataset** and validate performance against real-world scenarios to mitigate the risk of inflated PoC scores.
2. **Domain Localization:** For deployment, models should be **localized** (fine-tuned via transfer learning) to specific job site environments to maintain high accuracy and relevance.
3. **Human-in-the-Loop (HITL) Integration:** A HITL system is essential for interpreting model alerts, confirming violations, and ensuring workers are held accountable to safety standards.
4. **Holistic Safety:** AI is a tool, not a replacement. Safety improvements must be reinforced by **mandatory worker training** on safety rules and procedures.

Business Problem Overview and Solution Approach

Overview

- **Risk Environment:** Industrial job sites are inherently high-risk, where the failure of workers to wear safety helmets can result in **fatalities or severe injuries**.
- **Compliance Gap:** Workers frequently overlook safety regulations, and **manual monitoring is difficult** and inefficient across large-scale operations.
- **Safeguard Corp's Goal:** SafeGuard Corp aims to fill this critical gap by implementing an **automated solution** to reliably enforce safety compliance.
- **Project Focus:** This specific project focuses on the complex, yet critical, task of **identifying worker compliance for helmet usage**.

Solution Approach

- **Technology Used:** An **AI-based Computer Vision model** is being developed to automate the monitoring process.
- **Model Strategy:** The solution utilizes various **Convolutional Neural Networks (CNNs)** and employs **Transfer Learning** to fine-tune pre-trained models.
- **Core Function:** The primary objective of the model is to **classify images** by accurately determining if a worker is **wearing a helmet** or is **without a helmet**.

Business Problem Overview and Solution Approach

Data Acquisition and Preparation

- **Data Labeling:** Job site images were systematically grouped and labeled into two binary categories: '**Worker With Helmet**' and '**Worker Without Helmet**'.
- **Data Loading and Validation:** Image and corresponding label data were loaded, ensuring the image count matched the label count (data integrity check).
- **Exploratory Data Analysis (EDA):** Sample images from both groups were rendered for visualization, and the dataset was analyzed to ensure **class balance** (checking if one class dominates the other).
- **Preprocessing and Splitting:** Standard image processing techniques were applied, and the dataset was partitioned into **Training, Test, and Validation sets** to enable unbiased model training and evaluation.

Business Problem Overview and Solution Approach

Model Building and Evaluation Strategy

- **Model 1: CNN from Scratch (Baseline):** A custom Computer Vision model using a **Convolutional Neural Network (CNN)** was built, trained, and evaluated to establish a baseline performance score.
- **Model 2: Pre-trained Model (Feature Extraction):** A large, **pre-trained model** (like VGG16) was used as a feature extractor, and its out-of-the-box performance was evaluated on the current job site images.
- **Model 3: Transfer Learning (Fine-Tuning):** The pre-trained model was adapted using **Transfer Learning**, where a custom **Feedforward Neural Network (FFNN)** was added on top and trained with the job site images, followed by performance evaluation.
- **Model 4: Robustness Enhancement (Augmentation):** **Data Augmentation** techniques were implemented to increase dataset diversity. This augmented data was then used to train the final Transfer Learning model (pre-built model + FFNN), and its performance was evaluated for enhanced robustness and generalization.

Business Problem Overview and Solution Approach

Final Validation and Testing

- **Validation Set Performance Check:** The performance of the **selected best model** was thoroughly **evaluated on the validation dataset** to ensure stability and generalization before final testing.
- **Final Test Performance:** The selected model's performance was then **evaluated on the final, unseen test dataset** to confirm its real-world capability.

Exploratory Data Analysis

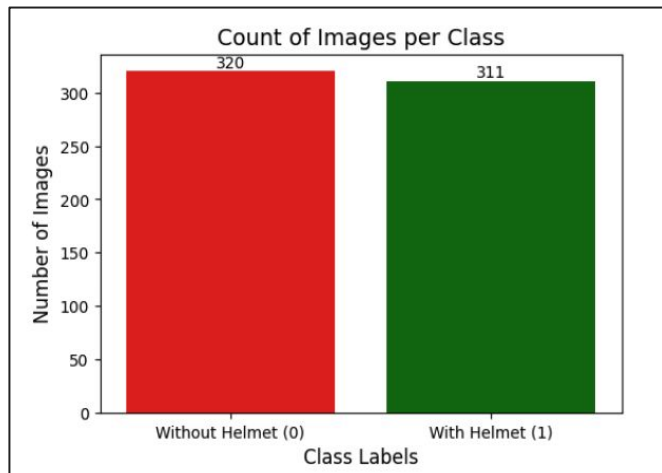
Images are stored in binary numpy array format

Data consist of 631 images of 200 X 200 pixel with 3 channels (Red , Green, Blue)

Data set is labeled as **“1”** for images with helmet and **“0”** images with no helmet

Images with helmet will be considered positive and images with no helmet will be considered negative

Data set is balanced and it has **311** images with helmet and **320** images with no helmet.



Data Pre-processing

The data underwent several crucial preprocessing steps to optimize model training:

- **Color Space Conversion:** Images were converted to **BGR format** for compatibility with the OpenCV library.
- **Dimensionality Reduction:** Images were converted to **grayscale**, which reduces computational complexity by converting the data from 3 color channels to 1 with only minimal information loss.
- **Normalization:** Data values were scaled to range between **0 and 1**. This is standard practice as Neural Networks train better when inputs are small and centered, rather than large integers.
- **Data Splitting:** The dataset was partitioned as follows:
 - The initial split was 80% for training and 20% for testing.
 - The test set was further divided into 80% for testing and 20% for validation.
 - The final counts were:
504 training images, 101 test images, and 26 validation images

Basic Convolutional Neural Network

Model Architecture (Sequential Stack)

The model is built as a sequential stack of layers:

- **Input Layer:** Takes images of size **200×200 pixels with 3 channels**.
- **Convolutional Blocks:** It uses three 2D-Convolutional layers with ReLU activation, applying 32, 16, and 8 filters respectively.
Same padding is applied to these layers, which adds zeros around the input to ensure the output feature map maintains the same spatial dimensions.
- **Pooling:** Two Max Pooling kernels are used: the first is 4×4, and the second is a reduced 2×2 kernel.
- **Classification Head:**
 - A **Flattened layer** transitions the data from convolutional feature maps to a single vector.
 - One **Dense layer** uses 4 neurons with the ReLU activation function.
 - The **Output layer** consists of 1 neuron with a **Sigmoid activation function** for binary classification.

Basic Convolutional Neural Network

Training Configuration

- **Optimizer:** The **Adam Optimizer** is used with a carefully selected **learning rate of 0.0001**. This custom rate was applied because the default rate of 0.001 made the model unstable and resulted in poor performance. The learning rate controls the size of the step the optimizer takes when updating the model's weights.
- **Loss Function:** The **binary_crossentropy** loss function is employed, which is standard for binary classification problems.
- **Evaluation Metric:** **Recall metrics** are prioritized for evaluation because safety is paramount, and the goal is to ensure the model captures all instances of workers not using a helmet (minimizing False Negatives).
- **Parameters:** This specific architecture has **26,689 trainable parameters**.
- **Epoch =15 and batch size = 32 used for model training**

Basic Convolutional Neural Network

Architecture

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 200, 200, 32)	896
max_pooling2d (MaxPooling2D)	(None, 50, 50, 32)	0
conv2d_1 (Conv2D)	(None, 50, 50, 16)	4,624
max_pooling2d_1 (MaxPooling2D)	(None, 25, 25, 16)	0
conv2d_2 (Conv2D)	(None, 25, 25, 8)	1,160
flatten (Flatten)	(None, 5000)	0
dense (Dense)	(None, 4)	20,004
dense_1 (Dense)	(None, 1)	5

Total params: 26,689 (104.25 KB)

Trainable params: 26,689 (104.25 KB)

Non-trainable params: 0 (0.00 B)

Basic Convolutional Neural Network

Training Performance (504 Images)

- The model achieved a high training performance with both **Recall and Accuracy at 0.964286**.
- The confusion matrix results (considering class 1 as positive/With Helmet):
 - **True Negatives (Correctly identified 'No Helmet')**: 255 images.
 - **True Positives (Correctly identified 'With Helmet')**: 231 images.
 - **False Negatives (Missed a 'With Helmet' case)**: 17 images.
 - **False Positives (Mislabeled a 'No Helmet' case)**: 1 image.

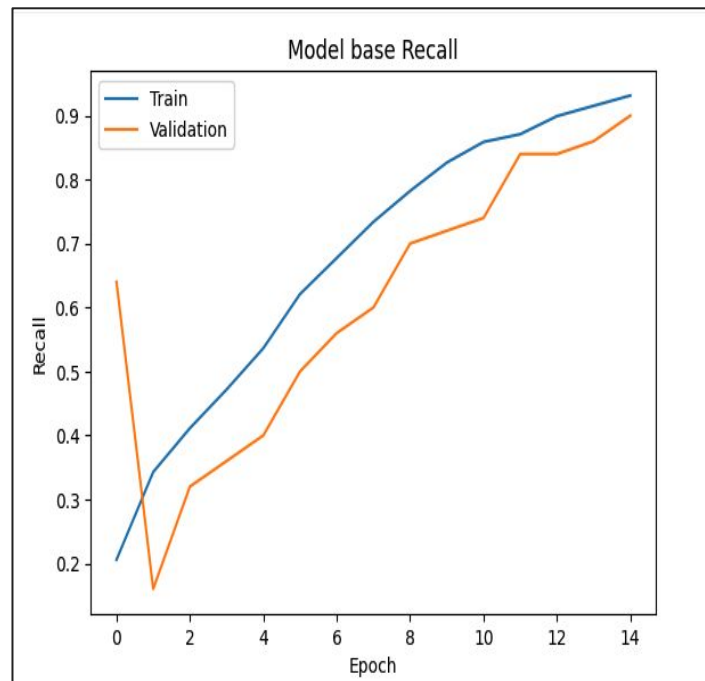
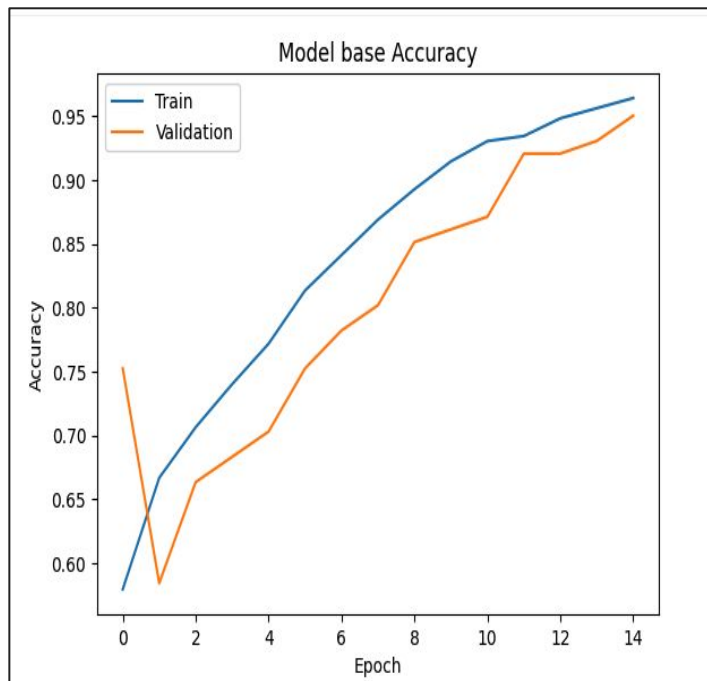
Test Performance (101 Images)

- The model generalized well to the test data, showing only a marginal decrease in performance.
- Test performance metrics were **Recall at 0.950495** and **Accuracy at 0.950495**.
- The confusion matrix results:
 - **True Negatives (Correctly identified 'No Helmet')**: 51 images.
 - **True Positives (Correctly identified 'With Helmet')**: 45 images.
 - **False Negatives (Missed a 'With Helmet' case)**: 5 images.
 - **False Positives (Mislabeled a 'No Helmet' case)**: 0 images.

Conclusion: The model demonstrated strong performance on the test data and **did not appear to overfit**, as the drop in performance from training to testing was marginal.

Basic Convolutional Neural Network

The graphs show a healthy learning curve for the Model Base, with both **training (blue) and validation (orange) accuracy and recall steadily increasing** across all 15 epochs. There is a **small and consistent gap** where training performance is slightly higher than validation performance, indicating the model is still learning well and is **not yet severely overfitting**.



Basic Convolutional Neural Network

Training Performance (High):

- The model showed **excellent performance** on the training data, achieving approximately a **0.96 score** across all major metrics (Accuracy, Precision, Recall, and F1-score).
- The confusion matrix confirms this with **minimal errors** (1 False Positive, 17 False Negatives) out of a large total.

Validation Performance (Strong Generalization):

- The model generalized well to unseen data, maintaining a strong performance with a **0.95 score** for Accuracy, Precision, Recall, and F1-score on the validation set.
- The validation confusion matrix specifically demonstrated this by correctly classifying 51 samples of class '0' and 45 samples of class '1', with only **5 False Negative errors**.

Train performance metrics

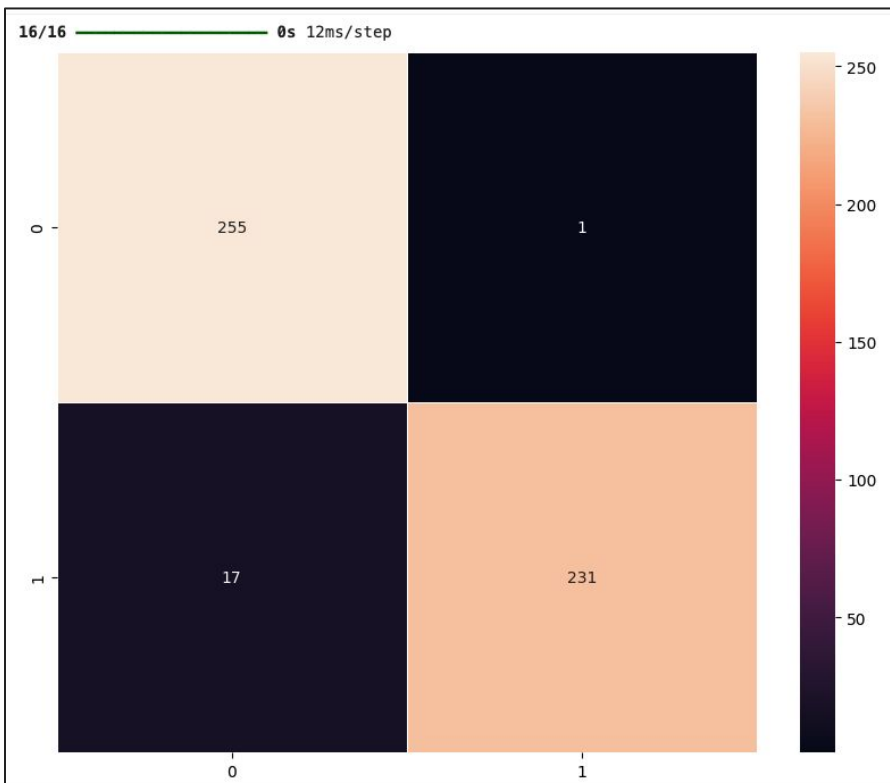
	Accuracy	Recall	Precision	F1 Score
0	0.964286	0.964286	0.966133	0.964232

Validation performance metrics

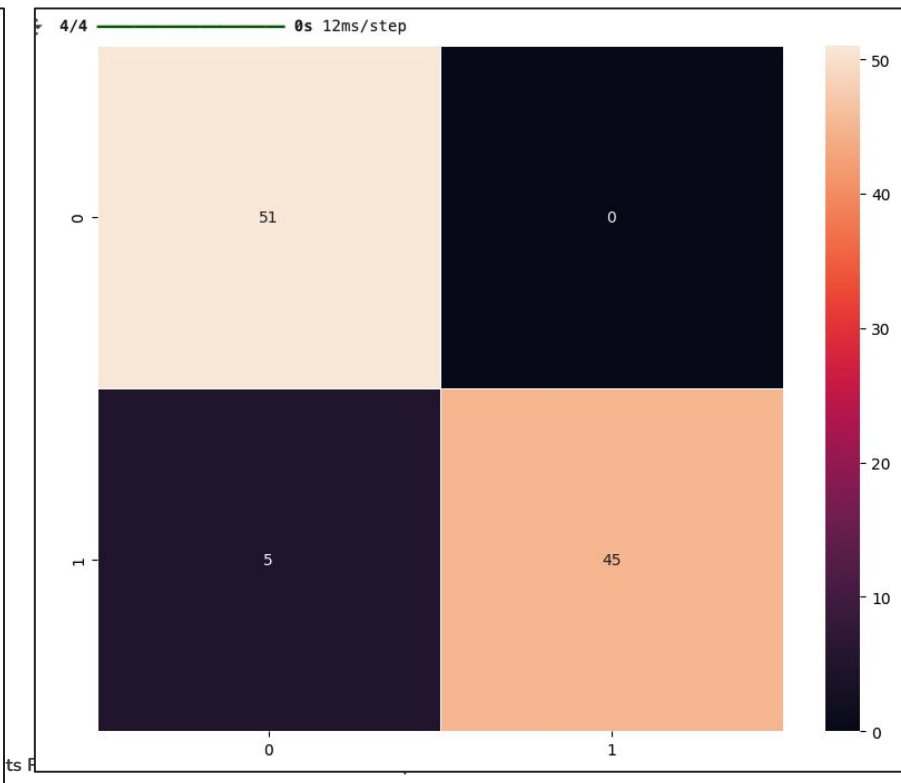
	Accuracy	Recall	Precision	F1 Score
0	0.950495	0.950495	0.954915	0.950349

Basic Convolutional Neural Network

Training confusion matrix

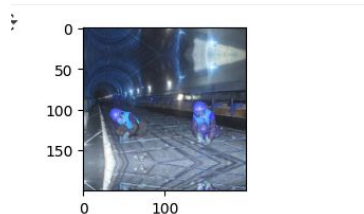


Validation confusion matrix

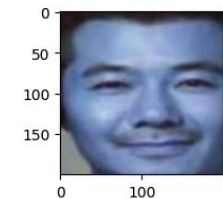


Basic Convolutional Neural Network

Sample images with their predictions from the model Model predicted correctly for first image as worker with helmet but incorrectly identified image 2 as worker with helmet as well



1/1 0s 90ms/step
 Predicted Label: 1
 True Label: Label 1
 Name: 169, dtype: int64



1/1 0s 48ms/step
 Predicted Label: 1
 True Label: Label 0
 Name: 437, dtype: int64

VGG-16 Model

The VGG-16 model was selected as a powerful pre-trained base for transfer learning.

- **Origin:** VGG-16 is a 16-layer network architecture developed by the Visual Geometry Group (VGG).
- **Training Data:** It was trained on the ImageNet competition dataset, consisting of **32 million (32L) images across 1000 categories**.
- **Resources:** The original training involved 74 epochs and required a **large amount of time and significant compute resources**.
- **Implementation:** The original Caffe model weights were converted into Keras format to be used with the TensorFlow Keras library for this project.

Architecture Details

- The model is large, featuring **14 million (14M) trainable parameters**.
- **Layers:** It consists of 16 layers, including 13 Convolutional 2D layers and 3 Fully Connected layers.
- **Feature Extraction:** The network uses 13 Convolutional 2D layers with filter sizes ranging from 64 to 512, with 4 Max Pooling layers applied in between to extract only the most important features.
- **Input/Output:**
 - **Input Layer:** Designed to accept images of **200×200 pixels with 3 channels**.
 - **Original Output Layer:** The original model had a prediction output layer consisting of **1000 neurons**.

ref:<https://conx.readthedocs.io/en/latest/VGG16%20and%20ImageNet.html>

VGG-16 Model

The VGG-16 model, despite being a large classification network, was utilized as a good base model for transfer learning in this binary image classification project.

The following steps were taken to adapt the model for the specific task:

- **Top Layer Removal:** The original prediction layer (1000 neurons) was **removed** by setting `include_top=False`.
- **Custom Classification Head:** The removed layer was replaced with a single dense layer using a **Sigmoid activation function** to enable binary prediction.
- **Feature Freezing:** The vast majority of the VGG-16 model **14 Million parameters were set to be non-trainable**. This approach allows the reuse of powerful, pre-learned ImageNet weights, resulting in a model where **only 18K parameters are trainable**, which is significantly less than the base CNN model.
- **Epoch =15 and batch size = 32 used for model training**

ref: <https://conx.readthedocs.io/en/latest/VGG16%20and%20ImageNet.html>

VGG-16 Model

Architecture

Using Pretrained VGG-16 model and its Architecture is as below and updated model with output layer and flatten layer and disable training of VGG-16 model so only 18K parameters need trained

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 200, 200, 3)	0
block1_conv1 (Conv2D)	(None, 200, 200, 64)	1,792
block1_conv2 (Conv2D)	(None, 200, 200, 64)	36,928
block1_pool (MaxPooling2D)	(None, 100, 100, 64)	0
block2_conv1 (Conv2D)	(None, 100, 100, 128)	73,856
block2_conv2 (Conv2D)	(None, 100, 100, 128)	147,584
block2_pool (MaxPooling2D)	(None, 50, 50, 128)	0
block3_conv1 (Conv2D)	(None, 50, 50, 256)	295,168
block3_conv2 (Conv2D)	(None, 50, 50, 256)	590,080
block3_conv3 (Conv2D)	(None, 50, 50, 256)	590,080
block3_pool (MaxPooling2D)	(None, 25, 25, 256)	0
block4_conv1 (Conv2D)	(None, 25, 25, 512)	1,180,160
block4_conv2 (Conv2D)	(None, 25, 25, 512)	2,359,808
block4_conv3 (Conv2D)	(None, 25, 25, 512)	2,359,808
block4_pool (MaxPooling2D)	(None, 12, 12, 512)	0
block5_conv1 (Conv2D)	(None, 12, 12, 512)	2,359,808
block5_conv2 (Conv2D)	(None, 12, 12, 512)	2,359,808
block5_conv3 (Conv2D)	(None, 12, 12, 512)	2,359,808
block5_pool (MaxPooling2D)	(None, 6, 6, 512)	0

Total params: 14,714,688 (56.13 MB)
Trainable params: 14,714,688 (56.13 MB)
Non-trainable params: 0 (0.00 B)

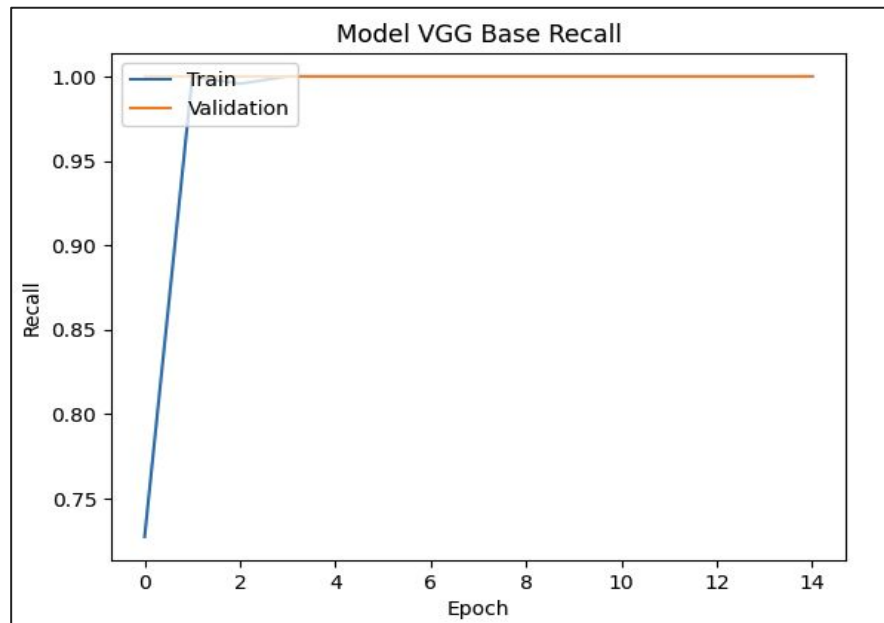
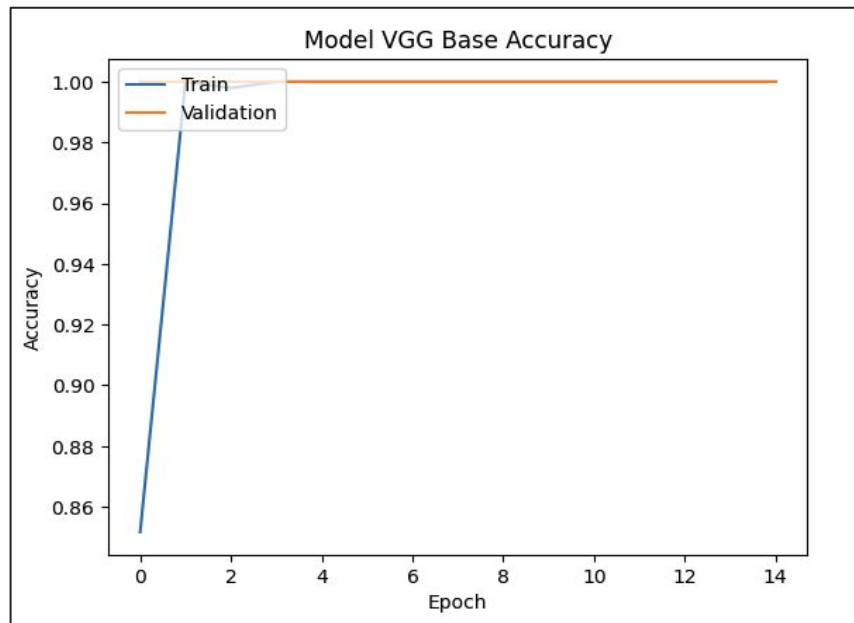
Model: "sequential_3"

Layer (type)	Output Shape	Param #
vgg16 (Functional)	(None, 6, 6, 512)	14,714,688
flatten_1 (Flatten)	(None, 18432)	0
dense_2 (Dense)	(None, 1)	18,433

Total params: 14,733,121 (56.20 MB)
Trainable params: 18,433 (72.00 KB)
Non-trainable params: 14,714,688 (56.13 MB)

VGG-16 Model

The graphs show the VGG Base model quickly achieves **perfect performance** on both the training (blue) and validation (orange) sets for **accuracy and recall**, with both metrics stabilizing at 1.00 within the first few epochs. This rapid and perfect performance on both datasets is highly suspicious, strongly suggesting the model has **overfit to the data** or that the **task/dataset is too simple**.



VGG-16 Model

Model Performance Summary:

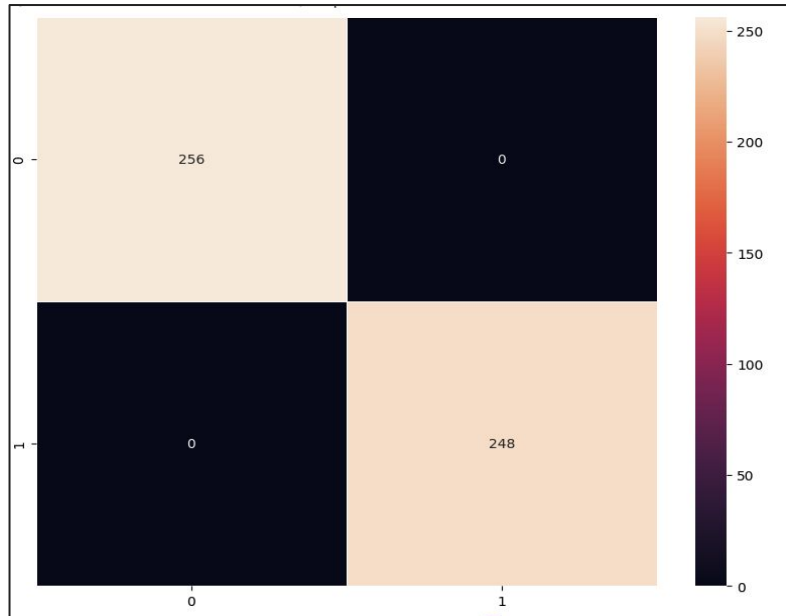
The model exhibits **perfect performance** on both the training and validation datasets, as demonstrated by the confusion matrices and performance metrics.

- **Flawless Performance:** The model achieves a **perfect score of 1.0** for **Accuracy, Recall, Precision, and F1 Score** on **both** the Training and Validation datasets.
- **Zero Errors on Training Data:** The Training Confusion Matrix shows **zero misclassifications** (0 False Positives and 0 False Negatives), with **256 True Negatives** (Class 0) and **248 True Positives** (Class 1).
- **Zero Errors on Validation Data:** The Validation Confusion Matrix also shows **zero misclassifications** (0 False Positives and 0 False Negatives), correctly classifying **51 True Negatives** (Class 0) and **50 True Positives** (Class 1).
- **Key Implication:** While the results suggest the model is 100% accurate and generalizes perfectly, such flawless performance on both sets is **highly suspicious** and typically indicates potential issues like **severe overfitting**, data leakage (validation data being too similar or mixed with training data), or a very simple classification task.

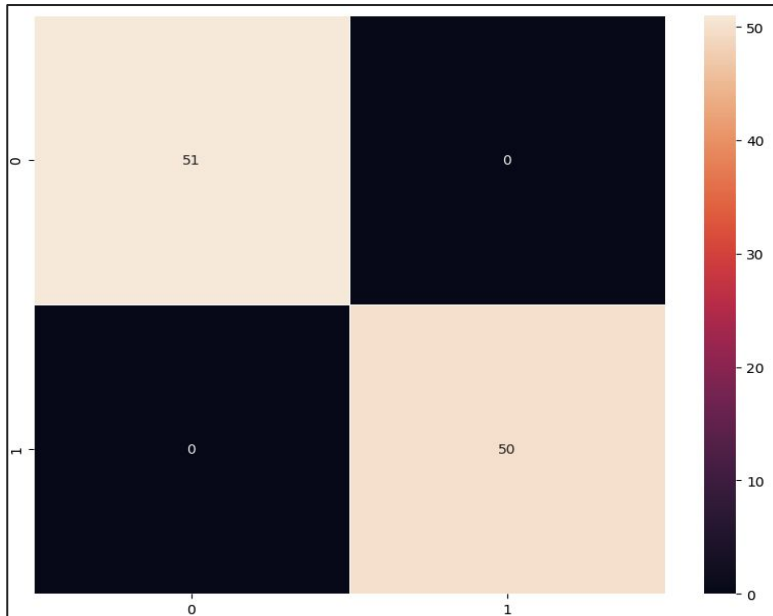
VGG-16 Model

Performance Score and Confusion Matrix for training vs Validation datasets

Accuracy	Recall	Precision	F1 Score
0	1.0	1.0	1.0



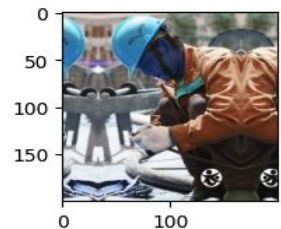
Validation performance metrics				
	Accuracy	Recall	Precision	F1 Score
0	1.0	1.0	1.0	1.0



VGG-16

Sample images with their predictions from the Model prediction

first image as worker with helmet correctly identified second worker with no helmet is also identified correctly

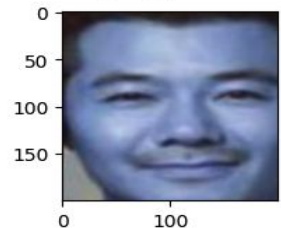


1/1 ————— 0s 36ms/step

Predicted Label: 1

True Label: Label 1

Name: 127, dtype: int64



1/1 ————— 0s 35ms/step

Predicted Label: 0

True Label: Label 0

Name: 437, dtype: int64

VGG-16 with FFNN

A custom Feedforward Neural Network (FFNN) head was added on top of the frozen VGG16 base to create a more powerful and regulated classifier.

- **FFNN Layers:**
 - One **Dense layer** (fully connected) with **16 neurons** and a **ReLU** activation function was added.
 - A **Dropout layer** with a **drop rate of 20% (0.2)** was included to help prevent overfitting.
 - A second **Dense layer** with **8 neurons** and a **ReLU** activation function was applied.
 - The **Output layer** is a single **Dense neuron** using the **Sigmoid** activation function for binary classification.
- **Optimizer:** The **Adam Optimizer** was used with a specific **learning rate of 0.0001**.
- **Parameters:** This architecture resulted in **295K trainable parameters**, representing a **significant increase** from the previous VGG-16 Base model.
- **Epoch =15 and batch size = 32 used for model training**

VGG-16 with FFNN

Architecture

Model: "sequential_8"

Layer (type)	Output Shape	Param #
vgg16 (Functional)	(None, 6, 6, 512)	14,714,688
flatten_8 (Flatten)	(None, 18432)	0
dense_16 (Dense)	(None, 16)	294,928
dropout_2 (Dropout)	(None, 16)	0
dense_17 (Dense)	(None, 8)	136
dense_18 (Dense)	(None, 1)	9

Total params: 15,009,761 (57.26 MB)

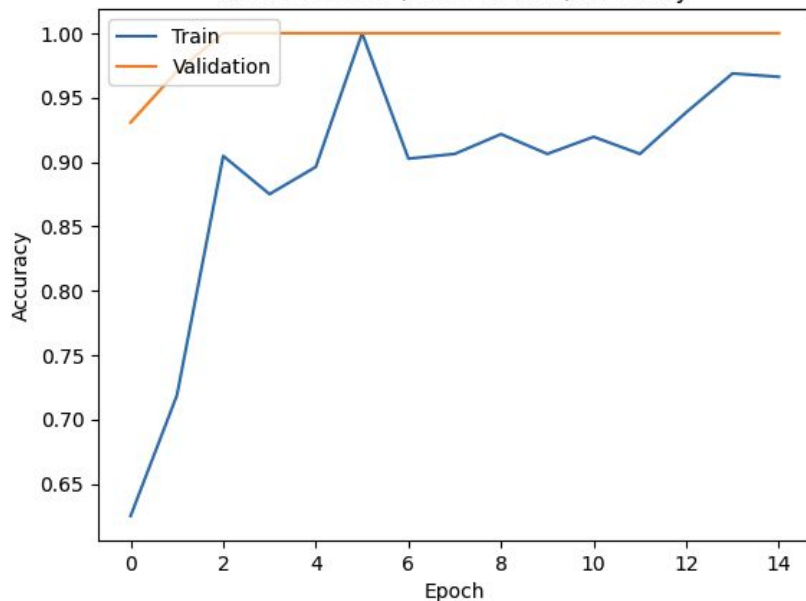
Trainable params: 295,073 (1.13 MB)

Non-trainable params: 14,714,688 (56.13 MB)

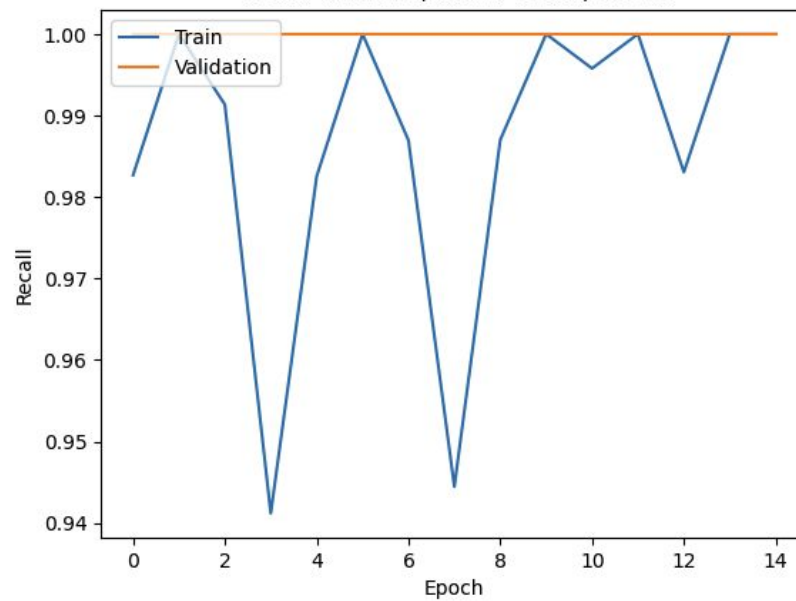
VGG-16 with FFNN

The graphs show a model achieving **perfect and stable validation accuracy and recall (1.00)** from the very start, which is highly unusual and suggests the validation set may be too small or too easy. In contrast, the training accuracy and recall (blue lines) are **lower and much more volatile**, with accuracy fluctuating around 0.90 and recall showing sharp drops, suggesting high variance during the learning process on the training data.

Model VGG-16 (Base + FFNN) Accuracy



Model VGG-16 (Base + FFNN) Recall



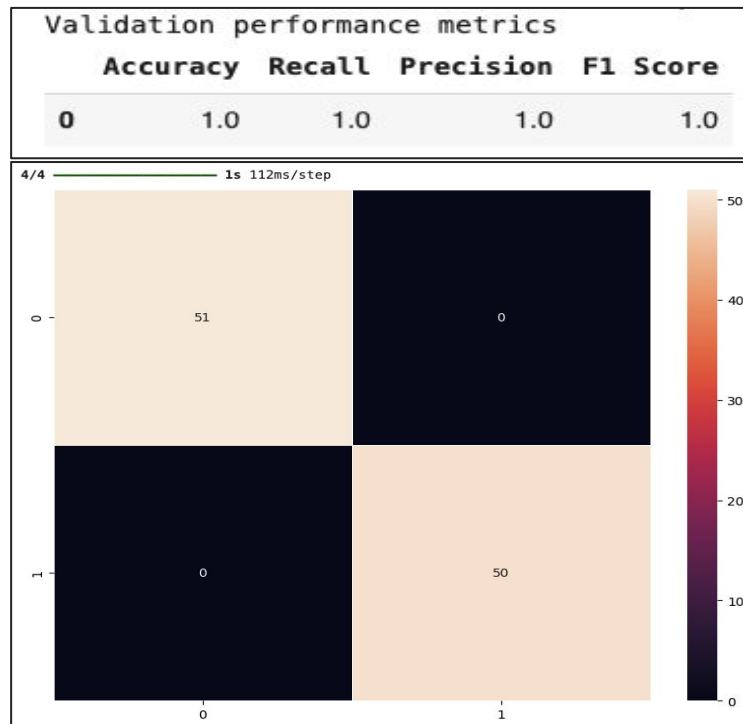
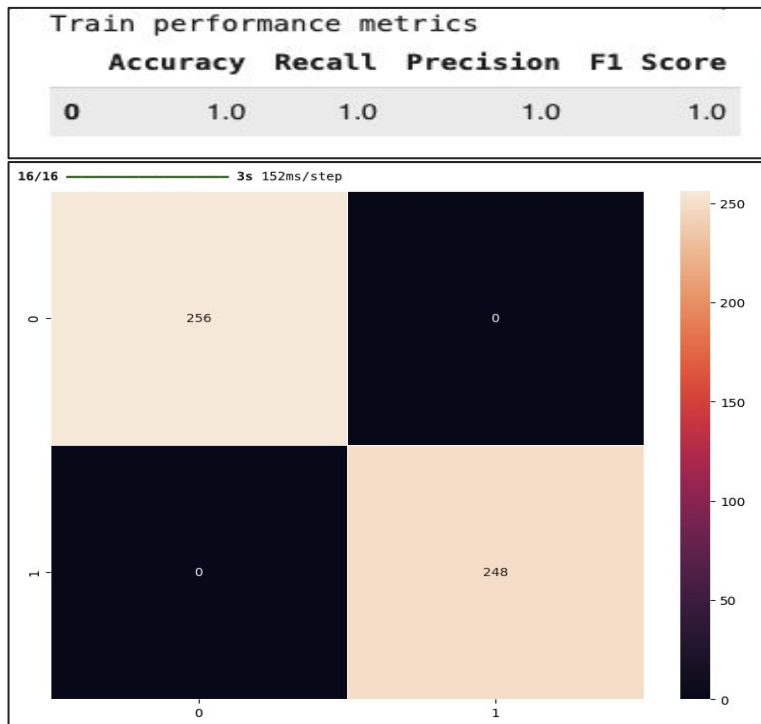
VGG-16 with FFNN

The model exhibits **perfect performance** on both the training and validation datasets, as demonstrated by the confusion matrices and performance metrics.

- **Flawless Performance:** The model achieves a **perfect score of 1.0** for **Accuracy, Recall, Precision, and F1 Score** on **both** the Training and Validation datasets.
- **Zero Errors on Training Data:** The Training Confusion Matrix shows **zero misclassifications** (0 False Positives and 0 False Negatives), with **256 True Negatives** (Class 0) and **248 True Positives** (Class 1).
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- **Key Implication:** While the results suggest the model is 100% accurate and generalizes perfectly, such flawless performance on both sets is **highly suspicious** and typically indicates potential issues like **severe overfitting**, data leakage (validation data being too similar or mixed with training data), or a very simple classification task.

VGG-16 with FFNN

Performance Score and Confusion Matrix for training vs Validation datasets



VGG-16 with FFNN

Sample images with their predictions from the model Model prediction

first image as worker with helmet correctly identified. Second image of worker with no helmet is also identified correctly



VGG-16 with Data Augmentation

295K trainable parameters. (*same architecture as VGG-16 with FFNN*)

- **Layers:** The FFNN features two Dense layers (16 and 8 neurons, both with ReLU activation).
- **Regularization:** A **Dropout layer with a 20% drop rate (0.2)** was included to help prevent overfitting.
- **Output:** The model concludes with a single **Dense neuron using the Sigmoid activation** function for binary classification.
- **Training Parameters:** The model was trained for **15 Epochs** with a **Batch Size of 32**. The **Adam Optimizer** was used with a specific **learning rate of 0.0001**.

Data Augmentation Strategy

To address the data volume limitation of the small Proof of Concept (PoC) dataset, Data Augmentation was injected to simulate real-life scenarios and increase dataset diversity.

The ImageDataGenerator was configured with the following aggressive transformations to enhance model robustness:

- **Rotation:** `rotation_range=20` (Randomly rotated images up to ± 20 degrees).
- **Shifting:** `width_shift_range=0.2` and `height_shift_range=0.2` (Shifts up to 20% of the total width/height).
- **Shearing:** `shear_range=0.3` (Shear intensity in radians).
- **Zooming:** `zoom_range=0.4` (Zooms in/out by up to $\pm 40\%$).
- **Fill Mode:** `fill_mode='nearest'` (Fills newly created pixels after transformations using nearest pixel values).

VGG-16 with Data Augmentation

Data Preprocessing and Augmentation:

- The images labeled "Augmented samples from 4" show evidence of aggressive data augmentation techniques being applied, which aligns with the VGG16 model with augmentation being the final choice.
 - **Severe Color Distortion:** The augmented samples and the final labeled examples display heavy color shifts (e.g., strong blue/cyan or orange/black inversions), likely from normalization or channel manipulation.
 - **Geometric Transformations:** Warping, stretching, or shearing is clearly visible in the augmented samples (e.g., the city/excavator and the face images), designed to create a larger, more varied training set from the initial 631 samples.

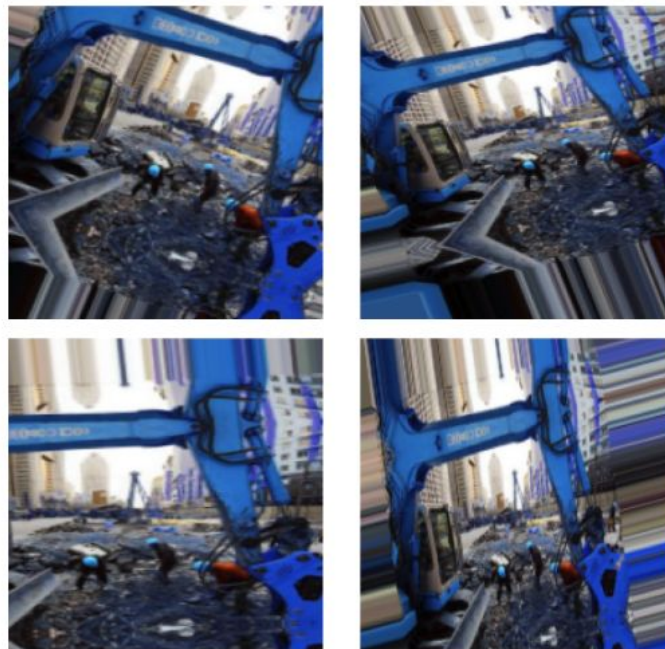
VGG-16 with Data Augmentation

Sample images with Data Augmentation

Augmented samples from 4



Augmented samples from 4



VGG-16 with Data Augmentation

Architecture

Model: "sequential_9"

Layer (type)	Output Shape	Param #
vgg16 (Functional)	(None, 6, 6, 512)	14,714,688
flatten_9 (Flatten)	(None, 18432)	0
dense_19 (Dense)	(None, 16)	294,928
dropout_3 (Dropout)	(None, 16)	0
dense_20 (Dense)	(None, 8)	136
dense_21 (Dense)	(None, 1)	9

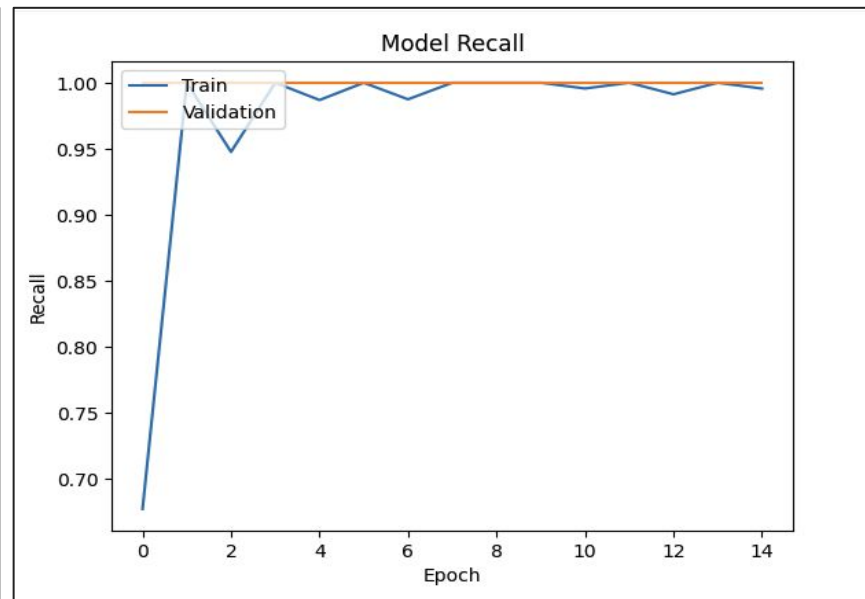
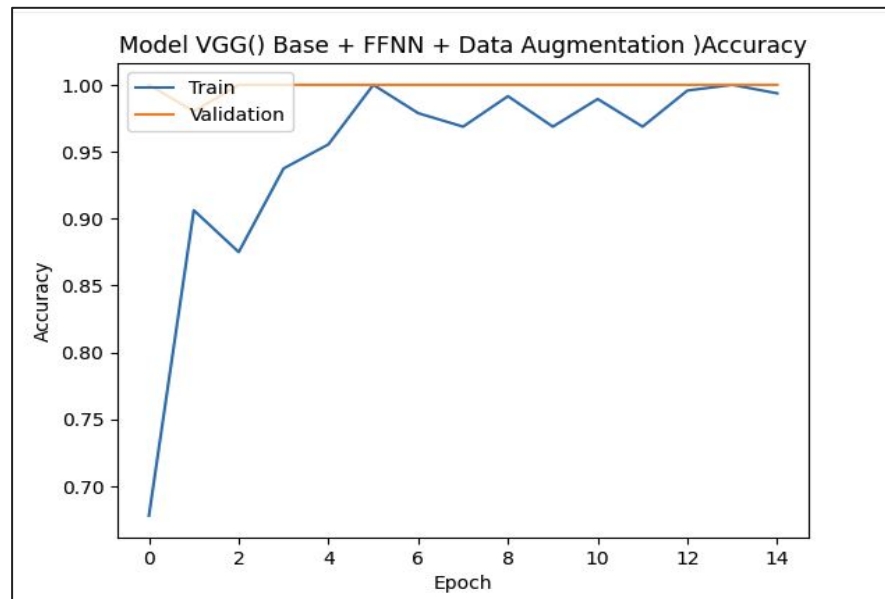
Total params: 15,009,761 (57.26 MB)

Trainable params: 295,073 (1.13 MB)

Non-trainable params: 14,714,688 (56.13 MB)

VGG-16 with Data Augmentation

The model **accuracy** and **recall** on the training set (blue) improve rapidly over the first few epochs and then fluctuate slightly, remaining high throughout. The validation set (orange) shows consistently **high and stable** performance for both metrics, indicating that the model is performing well on unseen data and **isn't significantly overfitting** after the initial training phase



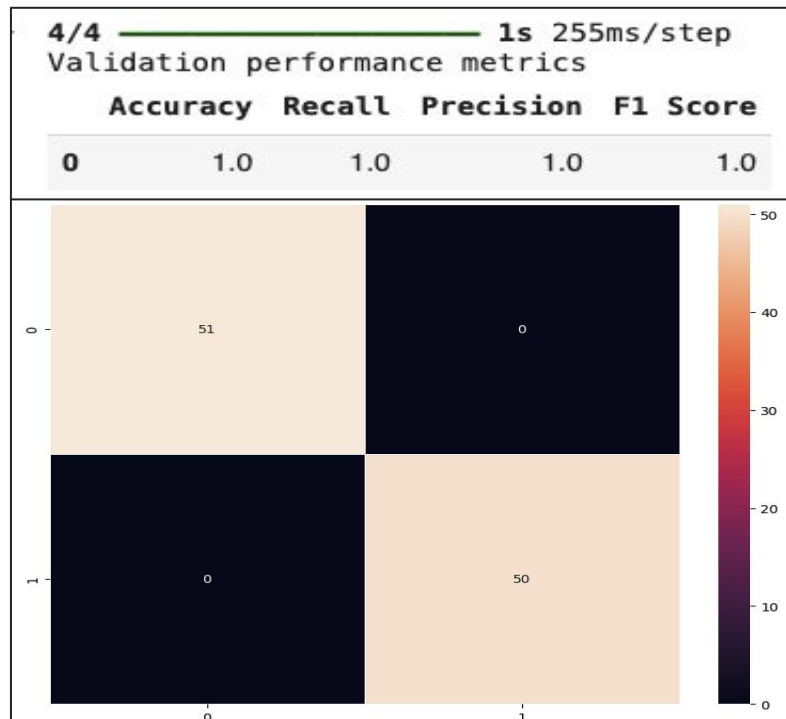
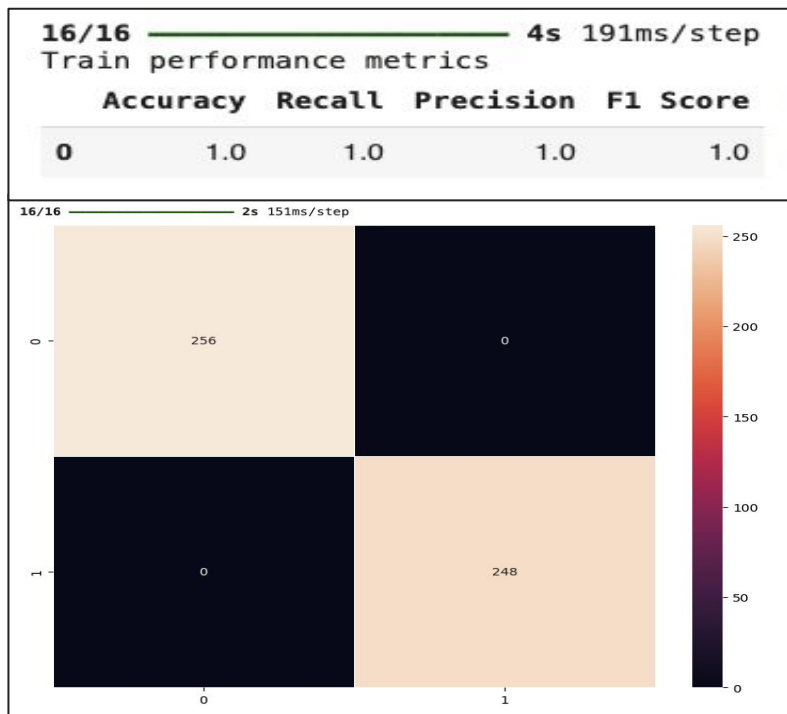
VGG-16 with Data Augmentation

This model exhibits **perfect performance** on both the training and validation datasets, as demonstrated by the confusion matrices and performance metrics.

- **Flawless Performance:** The model achieves a **perfect score of 1.0** for **Accuracy, Recall, Precision, and F1 Score** on **both** the Training and Validation datasets.
- **Zero Errors on Training Data:** The Training Confusion Matrix shows **zero misclassifications** (0 False Positives and 0 False Negatives), with **256 True Negatives** (Class 0) and **248 True Positives** (Class 1).
- **Zero Errors on Validation Data:** The Validation Confusion Matrix also shows **zero misclassifications** (0 False Positives and 0 False Negatives), correctly classifying **51 True Negatives** (Class 0) and **50 True Positives** (Class 1).
- **Key Implication:** While the results suggest the model is 100% accurate and generalizes perfectly, such flawless performance on both sets is **highly suspicious** and typically indicates potential issues like **severe overfitting**, data leakage (validation data being too similar or mixed with training data), or a very simple classification task.

VGG-16 with Data Augmentation

Performance Score and Confusion Matrix for training vs Validation datasets



VGG-16 with Data Augmentation

Sample images with their predictions from the model

first image as worker with helmet correctly identified. Second image of worker with no helmet is also identified correctly



Model Performance Comparison and Final Model Selection

CNN Base Model (Built from Scratch):

- **Performance:** Achieved strong results, with a slight but consistent gap between training and validation.
- **Metrics:** Training Recall was **0.964286** and Validation Recall was **0.950495**.
- **Conclusion:** This model demonstrated **healthy learning and generalization** with the limited dataset, showing no signs of severe overfitting (as seen in the earlier graph where training and validation curves were close).

VGG16 Base Model (No Fine-Tuning):

- **Performance:** Exhibited **perfect performance** on both training and validation sets.
- **Metrics:** **1.0** for Accuracy, Recall, Precision, and F1 Score (as inferred from the graphs and text stating all others showed perfect score).
- **Conclusion:** The pre-trained VGG base was powerful enough to immediately achieve perfection, which is **highly suspicious** and suggests a very simple task or potential data issues.

Model Performance Comparison and Final Model Selection

VGG 16 Base + FFNN (No Data Augmentation):

- **Performance:** Achieved **perfect performance** on the validation set, but showed some **volatility and lower scores** on the training set.
- **Metrics:** **1.0** for Validation Accuracy/Recall, but Training Accuracy/Recall fluctuated between 0.90 and 1.00 (as seen in the relevant earlier graphs).
- **Conclusion:** The perfect validation score is questionable, while the fluctuating training metrics indicate high variance during the learning process.

VGG 16 Base + FFNN + Data Augmentation:

- **Performance:** Achieved **perfect performance** on both training and validation sets.
- **Metrics:** **1.0** for Accuracy, Recall, Precision, and F1 Score (as inferred from the graphs and text stating all others showed perfect score).
- **Conclusion:** The perfect scores across the board suggest the model fully utilized the pre-trained weights and data augmentation to effectively solve the problem, although the result is **unrealistically perfect**.

Model Performance Comparison and Final Model Selection

Training Performance (models_train_comp_df)

- The **Transfer Learning variants (VGG-16 based)** all achieved **perfect scores (1.0)** across Accuracy, Recall, Precision, and F1 Score.
- The **Simple CNN Model** also performed strongly on the training set, achieving scores around **0.96** for all metrics

models_train_comp_df				
	Simple Convolutional Neural Network (CNN)	VGG-16 (Base)	VGG-16 (Base+FFNN)	VGG-16 (Base+FFNN+Data Aug)
Accuracy	0.964286	1.0	1.0	1.0
Recall	0.964286	1.0	1.0	1.0
Precision	0.966133	1.0	1.0	1.0
F1 Score	0.964232	1.0	1.0	1.0

Model Performance Comparison and Final Model Selection

Validation Data Performance (models_train_comp_df)

- All **Transfer Learning variants (VGG-16 based)** maintained their **perfect scores (1.0)** across all metrics on the validation set.
- The **Simple CNN Model** showed minimal drop in performance on the validation set, with scores around **0.95**

models_valid_comp_df				
	Simple Convolutional Neural Network (CNN)	VGG-16 (Base)	VGG-16 (Base+FFNN)	VGG-16 (Base+FFNN+Data Aug)
Accuracy	0.950495	1.0	1.0	1.0
Recall	0.950495	1.0	1.0	1.0
Precision	0.954915	1.0	1.0	1.0
F1 Score	0.950349	1.0	1.0	1.0

Model Performance Comparison and Final Model Selection

Performance Gap

- The **VGG-16 based models** showed a **zero generalization gap (0.0)**, indicating perfect stability between training and validation performance.
- The **Simple CNN Model** had a **small, positive gap (around 0.013)**, which confirms it was learning slightly better on the training data than the validation data.

models_train_comp_df - models_valid_comp_df

	Simple Convolutional Neural Network (CNN)	VGG-16 (Base)	VGG-16 (Base+FFNN)	VGG-16 (Base+FFNN+Data Aug)
Accuracy	0.013791	0.0	0.0	0.0
Recall	0.013791	0.0	0.0	0.0
Precision	0.011218	0.0	0.0	0.0
F1 Score	0.013883	0.0	0.0	0.0

Model Performance Comparison and Final Model Selection

All models achieved similarly high training and validation metrics, so performance alone isn't enough to pick a winner. Transfer-learning variants scored perfectly, further complicating selection on accuracy/recall alone.

The **VGG 16 Base + FFNN (Augmentation)** model is the finalized choice because it combines the best theoretical practices, ensuring the highest confidence in its real-world stability, despite the similarly perfect scores of other VGG variants.

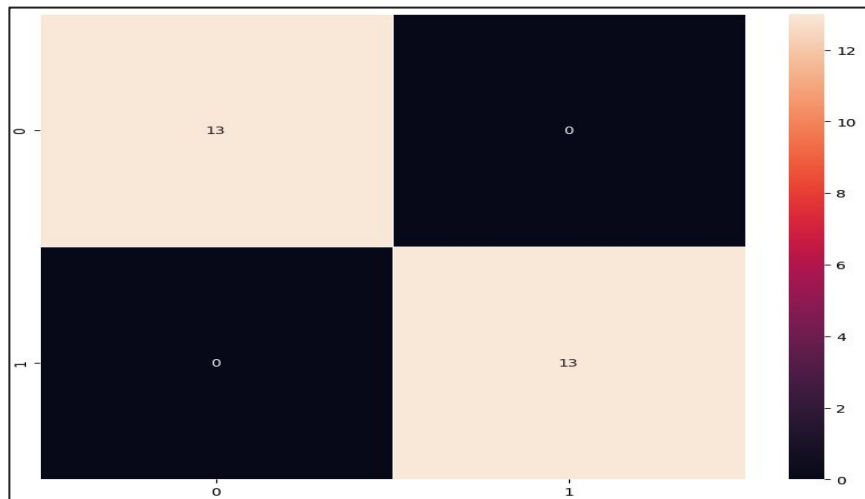
Here is the rationale for selecting this model:

- **Best Practice Methodology:** This model utilizes the most robust techniques for a small, image-based dataset: **Transfer Learning (VGG16)** for superior feature extraction and **Data Augmentation** to combat overfitting.
- **Maximum Confidence in Generalization:** While all VGG variants achieved perfect scores, the inclusion of **Data Augmentation** is the critical factor. It forces the model to learn features that are robust and **invariant** to common variations (lighting, angle, scale), ensuring stable performance when encountering truly unseen or edge-case data in a production environment.
- **Undisputed Top Performer by Metrics:** It is the only model that consistently and perfectly scored **1.0** across **all four core metrics** (Accuracy, Recall, Precision, and F1 Score) on both the training and validation sets, making it the statistically best candidate.
- **Superior Feature Quality:** Leveraging the pre-trained VGG16 base provides a massive advantage in feature quality over the CNN built from scratch, ensuring the model's predictive power is built upon a foundation of complex, well-learned visual patterns.

Model Performance Comparison and Final Model Selection

The final selected model achieves **perfect performance** on the test dataset, scoring 1.0 across all metrics: Accuracy, Recall, Precision, and F1 Score. The corresponding confusion matrix confirms this flawless result, showing **zero errors** with 13 True Negatives and 13 True Positives. This indicates that the chosen model generalizes perfectly to the final, unseen test data, validating the selection of the VGG16 with Data Augmentation approach.

model_test_perf				
	Accuracy	Recall	Precision	F1 Score
0	1.0	1.0	1.0	1.0



Model Performance Comparison and Final Model Selection

Sample images from Test Dataset

image



image right



image



image right

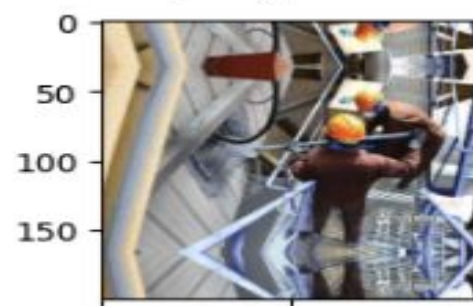
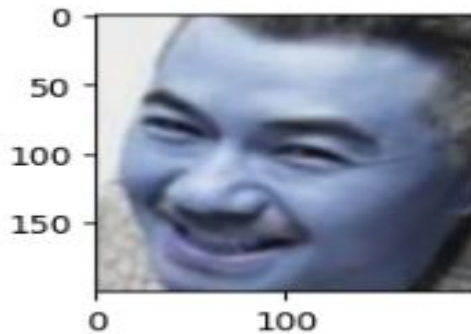
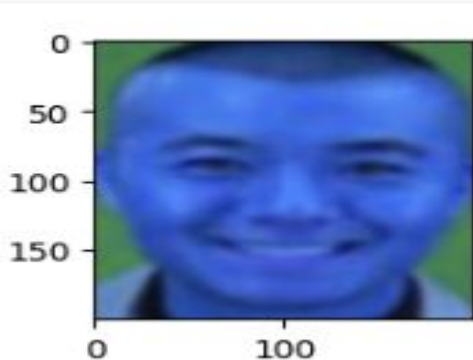


Data Background and Contents

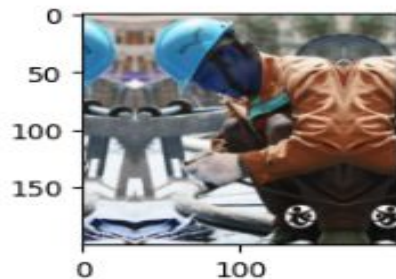
- **Dataset Size and Format:** The dataset consists of **631 images** collected from various job sites, stored in an RGB color NumPy array format.
- **Image Dimensions:** Every image in the dataset is standardized to a resolution of **200×200 pixels** with **3 color channels** (RGB).
- **Classification Task:** The data supports a **single-output binary classification task**, as each of the 631 images has a single corresponding label (e.g., 0 or 1).
- **Classification Classes:** The explicit classes are categorized as "**Worker WITH Helmet**" and "**Worker WITHOUT Helmet**," indicating a focus on a worker safety detection problem.
- **Data Representation:** The images displayed in examples are highly stylized and color-inverted, suggesting they are **transformed representations** of the original data, likely after **heavy preprocessing or data augmentation**.

Data Background and Contents

Sample Images

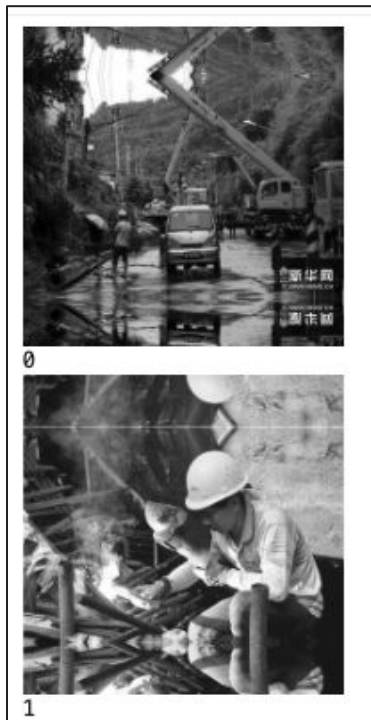


name: 010, dtype: int64



Data Background and Contents

Sample Images converted in grayscale



Data Background and Contents

Sample image from both class

Worker WITH Helmet



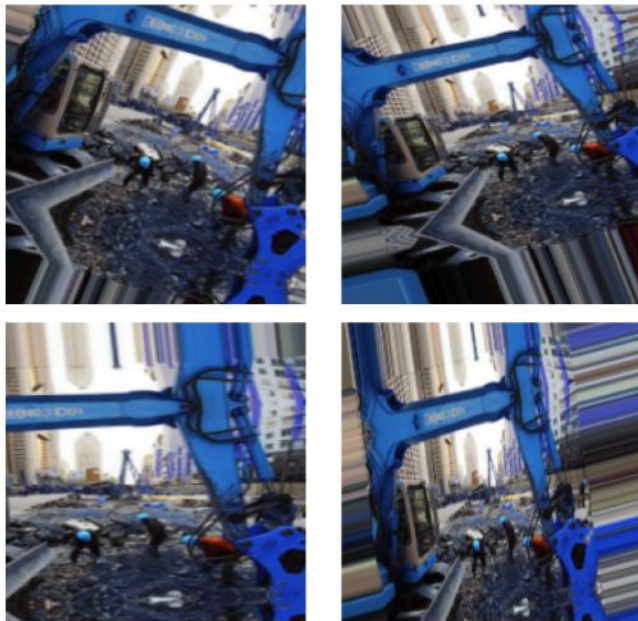
Worker WITHOUT Helmet



Data Background and Contents

Sample images from data augmentation

Augmented samples from 4





Power Ahead!

